

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

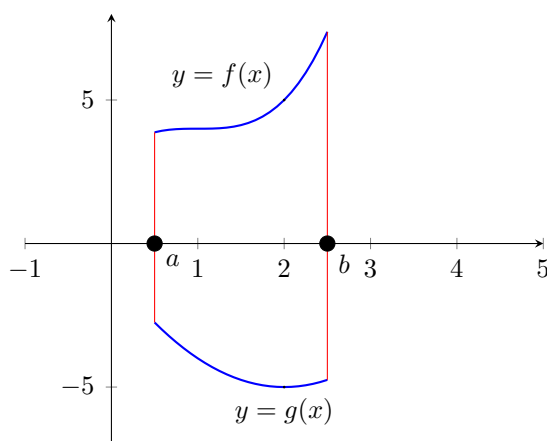
WEEK 13: LECTURE NOTE

CHAPTER 5. APPLICATIONS OF INTEGRATION

Area of a Region Between Two Intersecting Curves

Definition 1. If f and g are continuous on $[a, b]$ and $g(x) \leq f(x)$ for all x in $[a, b]$, then the area of the region bounded by the graphs of f and g and the vertical lines $x = a$ and $x = b$ is

$$\text{Area} = \int_a^b [f(x) - g(x)] dx.$$



Example 2. Find the area of the region bounded by the graphs of $y = x^2 + 2$, $y = -x$, $x = 0$ and $x = 1$.

Solution. In class.

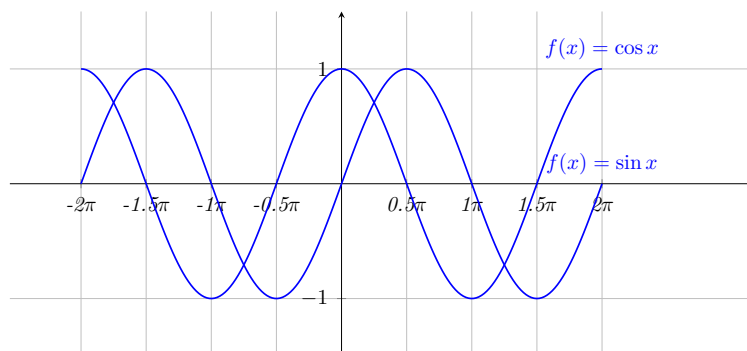
Area of a region between intersecting curves

A more common problem involves the area of a region bounded by two intersecting graphs, where the values of a and b must be calculated.

Example 3 (A Region Lying Between Two Intersecting Graphs). Find the area of the region bounded by the graphs of $f(x) = 2 - x^2$ and $g(x) = x$.

Solution. In class.

Example 4 (A Region Lying Between Two Intersecting Graphs). The sine and cosine curves intersect infinitely many times, bounding regions of equal areas, as shown in the Figure below. Find the area of one of these regions.

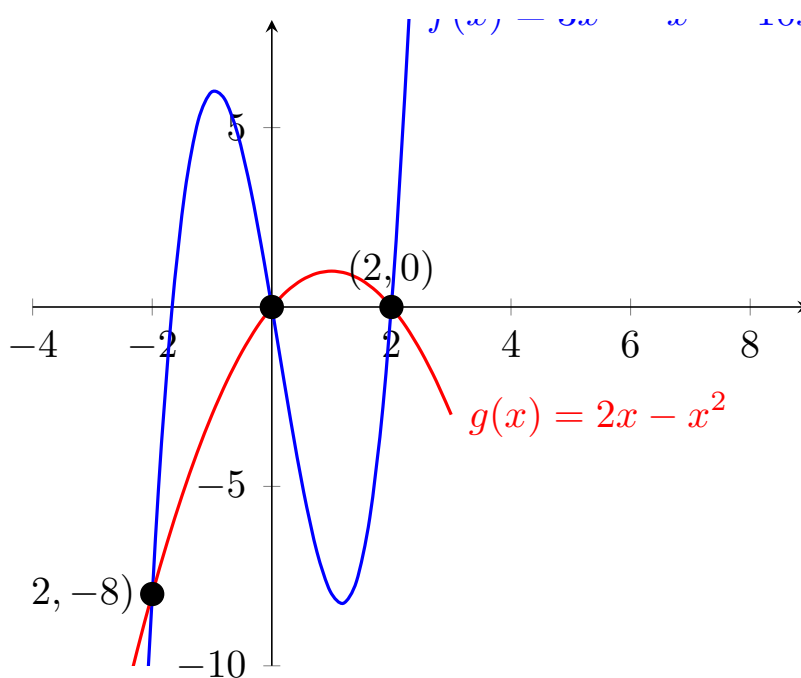


Solution. In class.

To find the area of the region between two curves that intersect at more than two points, first determine all points of intersection. Then check to see which curve is above the other in each interval determined by these points.

Example 5 (Curves That Intersect at More than Two Points). Find the area of the region between the graphs of

$$f(x) = 3x^3 - x^2 - 10x \quad \text{and} \quad g(x) = 2x - x^2$$



Solution. In class.

When the graph of a function of y is a boundary of a region, it is often convenient to use representative rectangles that are horizontal and find the area by integrating with respect to y . In general, to determine the area between two curves, you can use

$$Area = \int_{x_1}^{x_2} \underbrace{[(\text{top curve}) - (\text{bottom curve})]}_{\text{in variable } x} dx \quad \text{Vertical rectangles}$$

$$Area = \int_{y_1}^{y_2} \underbrace{[(\text{right curve}) - (\text{left curve})]}_{\text{in variable } y} dy \quad \text{Horizontal rectangles}$$

where (x_1, y_1) and (x_2, y_2) are either adjacent points of intersection of the two curves involved or points on the specified boundary lines.

Example 6 (Horizontal Representative Rectangles). *Find the area of the region bounded by the graphs of*

$$f(y) = 3 - y^2 \quad \text{and} \quad g(y) = y + 1.$$

Solution. In class.